

Numerical Methods

Chapter-2

Prepared by-
Md. Harun Or Rashid
Lecturer
Dept. of CSE, KiU

False Position Method

This is the oldest method for finding the real root of a nonlinear equation $f(x) = 0$. It is also known as Regula-Falsi or method of Chords. In this method, we choose two points a and b such that $f(a)$ and $f(b)$ are of opposite signs, then a root must lie in between a and b . Now the equation of the chord joining the two points $[a, f(a)]$ and $[b, f(b)]$ is given by-

$$\frac{y-f(a)}{x-a} = \frac{f(b)-f(a)}{b-a}$$

By taking the point of intersection on x-axis $[y=0]$ as an approximation to the root we obtain-

False Position Method (Cont'd)

$$\frac{0-f(a)}{x-a} = \frac{f(b)-f(a)}{b-a}$$

$$\Rightarrow x - a = -\frac{(b-a)f(a)}{f(b)-f(a)}$$

$$\Rightarrow x = a - \frac{(b-a)f(a)}{f(b)-f(a)}$$

$$\Rightarrow x = \frac{af(b)-bf(a)}{f(b)-f(a)}$$

Which is the first approximation to the root of $f(x) = 0$.

False Position Method (Cont'd)

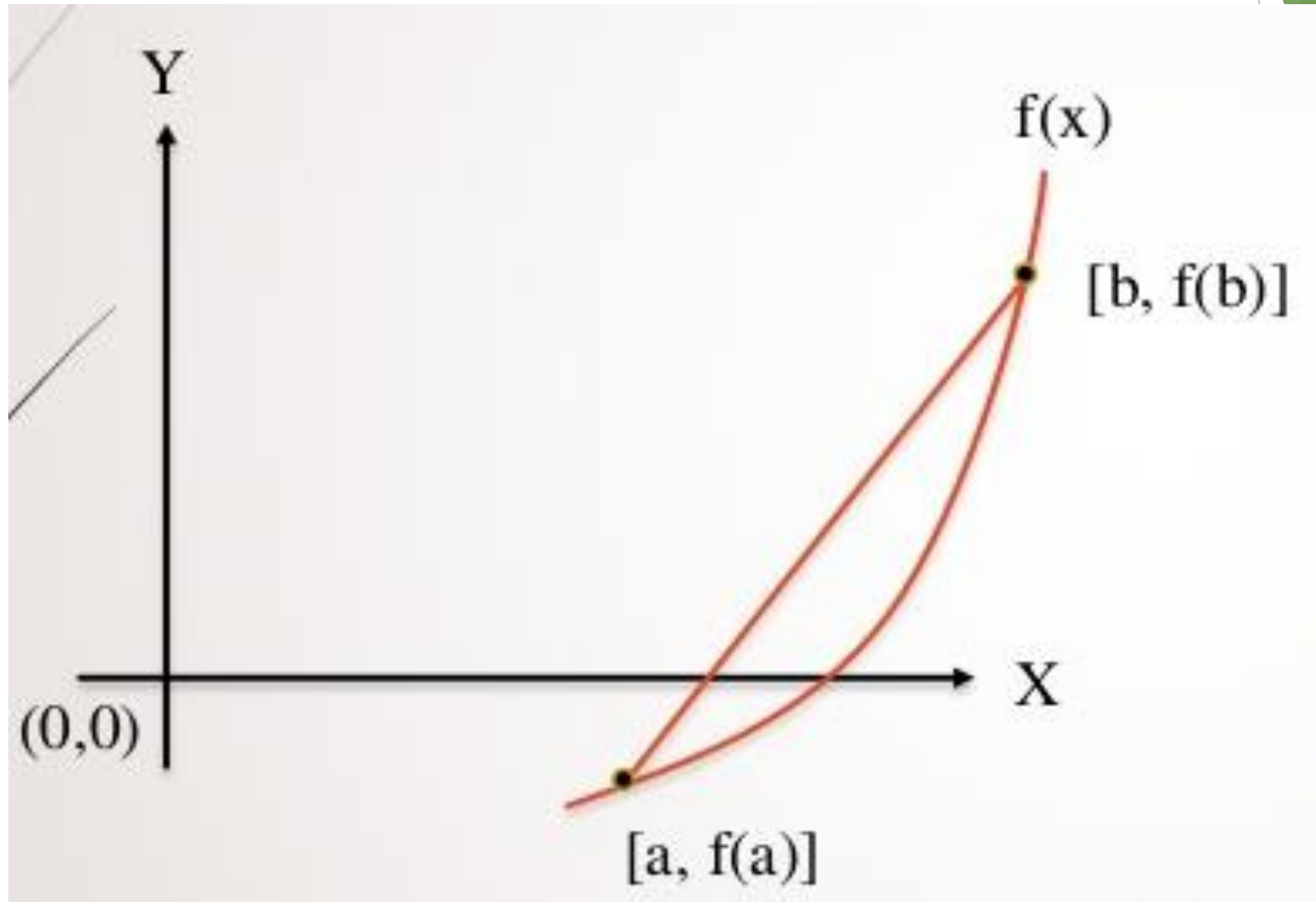


Fig: Graphical representation of the False Position method.

False Position Method (Cont'd)

It can be easily programmed using the following computational steps:

1. Choose two real numbers a and b such that $f(a)f(b) < 0$. [N.B.: a and b must be closer as much as possible]
2. Set $x_r = \frac{af(b)-bf(a)}{f(b)-f(a)}$.
3. (a) If $f(a)f(x_r) < 0$, the root lies in the interval (a, x_r) . Then, set $b = x_r$ and go to step 2 above.

(b) If $f(a)f(x_r) > 0$, the root lies in the interval (x_r, b) . Then, set $a = x_r$ and go to step 2 above.

(c) If $f(a)f(x_r) = 0$, it means that x_r is a root of the equation $f(x) = 0$ and the computation may be terminated.

False Position Method (Cont'd)

Example: Find a real root of the equation $x^3 - 2x - 5 = 0$ correct to three decimal places.

Solution: Given that,

$$x^3 - 2x - 5 = 0$$

Let, $f(x) = x^3 - 2x - 5$

Taking $a=2$ and $b=3$

$$\therefore f(2) = 2^3 - 2 \times 2 - 5 = -1 (< 0)$$

$$\therefore f(3) = 3^3 - 2 \times 3 - 5 = 16 (> 0)$$

Hence a root lies between 2 and 3.

False Position Method (Cont'd)

Step No.	a	b	$x_r = \frac{af(b) - bf(a)}{f(b) - f(a)}$	$f(x) = x^3 - 2x - 5$
1	2	3	2.058823529	-0.390799918
2	2.058823529	3	2.08126366	-0.147204064
3	2.08126366	3	2.089639211	-0.054676498
4	2.089639211	3	2.092739575	-0.020202859
5	2.092739575	3	2.093883709	-0.007450502
6	2.093883709	3	2.094305451	-0.002745669
7	2.094305451	3	2.094460845	-0.001011578

The difference between x_6 and x_7 is 0.000155394. Hence a root, correct to three decimal places is 2.094.

False Position Method(Cont'd)

Self Practise

Example- 2.7, 2.8, 2.9

N.B.: While computing trigonometric ratio, the specified value must be converted into degree using,

$$\text{Degree} = \frac{180 \times X}{3.14}$$

Where, X is in radian unit.

Task related to False position Method

1. Write some advantages of false position method.
2. Write some disadvantages of false position method.

Newton-Raphson Method

Let, x_0 be an approximate root of $f(x) = 0$ and let $x_1 = x_0 + h$ be the correct root so that $f(x_1) = 0$. Expanding $f(x_0 + h)$ by Taylor's series we obtain,

$$f(x_0) + hf'(x_0) + \frac{h^2}{2!}f''(x_0) + \dots \dots = 0$$

Neglecting the second and higher-order derivatives we have,

$$f(x_0) + hf'(x_0) = 0$$

$$\Rightarrow h = -\frac{f(x_0)}{f'(x_0)}$$

Newton-Raphson Method (Cont'd)

A better approximation than x_0 is therefore given by x_1 , where

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Successive approximation are given by $x_2, x_3, \dots, x_{n+1}$ where,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Which is the Newton-Raphson formula for real roots.

N.B.: Initial approximation x_0 is chosen sufficiently close to the correct root, δ . When δ is a multiple root, then Newton-Raphson method still converges but slowly.

Newton-Raphson Method (Cont'd)

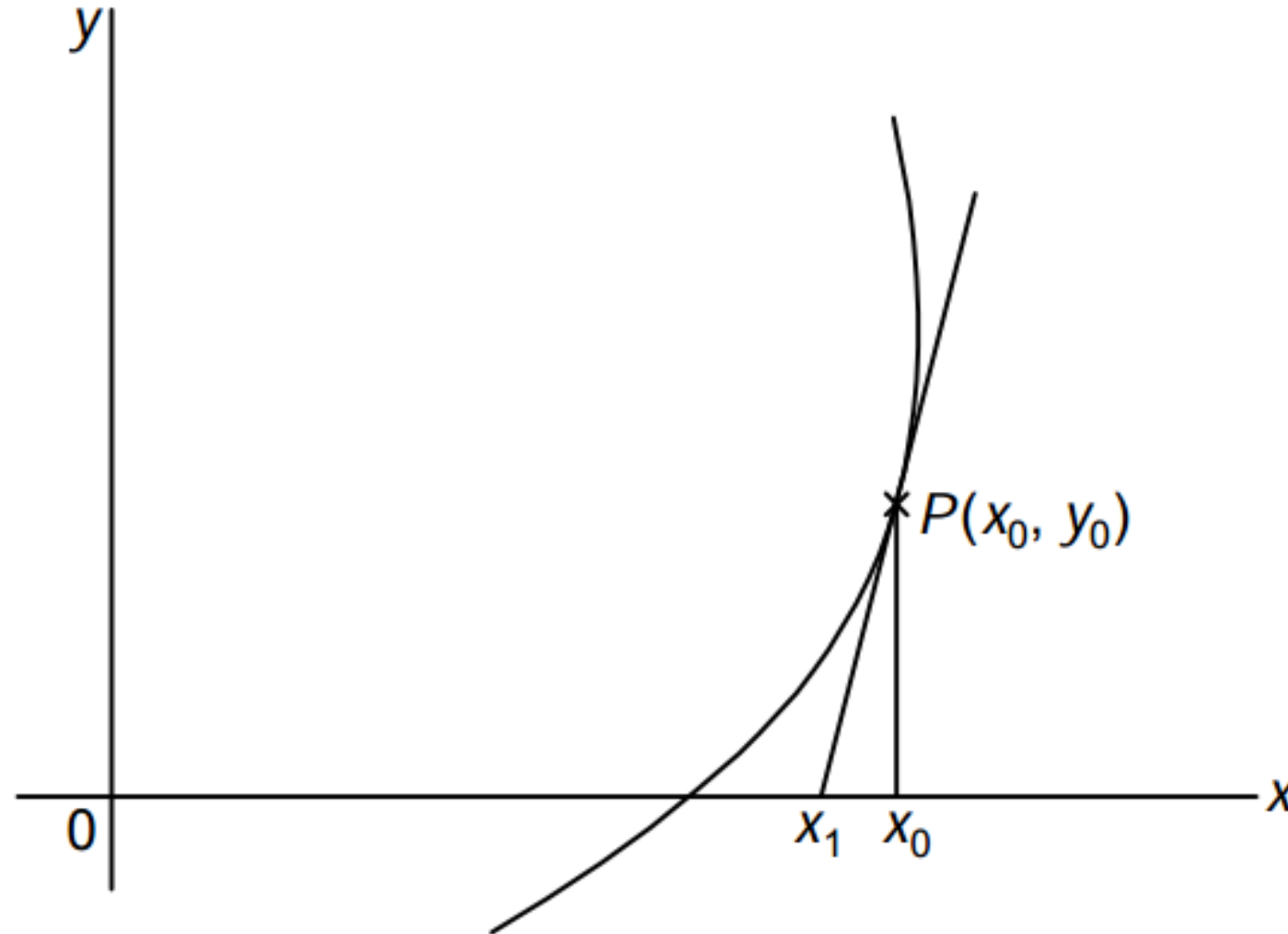


Fig: Graphical representation of the Newton-Raphson method.

Newton-Raphson Method (Cont'd)

Conditions for convergence of Newton-Raphson formula:

By Newton-Raphson formula we know,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \dots\dots\dots (i)$$

Equation (i) can be written as symbolically in the terms,

$$x_{n+1} = \varphi(x_n)$$

$$\therefore x = \varphi(x)$$

Since $\varphi(x_n) = x_{n+1}$ by iteration method for convergence

$$|\varphi'(x)| < 1$$

Newton-Raphson Method (Cont'd)

Here, $\varphi(x) = x - \frac{f(x)}{f'(x)}$

$$\therefore \varphi'(x) = 1 - \frac{f'(x) * f'(x) - f(x) * f''(x)}{(f'(x))^2}$$

$$= \frac{f(x) * f''(x)}{(f'(x))^2}$$

$$\therefore \left| \frac{f(x) * f''(x)}{(f'(x))^2} \right| < 1$$

Putting $x = x_n$, $\left| \frac{f(x_n) * f''(x_n)}{(f'(x_n))^2} \right| < 1$

This is the condition for convergence for a simple root.

Newton-Raphson Method (Cont'd)

Example-1: Use the Newton-Raphson method to find a root of the equation $x^3 - 2x - 5 = 0$.

Solution: Given that,

$$x^3 - 2x - 5 = 0$$

$$\text{Let, } f(x) = x^3 - 2x - 5$$

$$\therefore f'(x) = 3x^2 - 2$$

$$\text{Now, } x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2}$$

Newton-Raphson Method (Cont'd)

Choose $x_0 = 2$,

Step No.	x_n	$x_{n+1} = x_n - \frac{x_n^3 - 2x_n - 5}{3x_n^2 - 2}$
0	2	2.1
1	2.1	2.094568
2	2.094568	2.094551

The difference between x_2 and x_3 is 0.000017. Hence a root, correct to three decimal places is 2.094.

N.B.: Since Newton-Raphson method requires fewer iterations hence it converges more rapidly than previous methods. But it requires more computing time to evaluate two functions for each iteration.

Newton-Raphson Method (Cont'd)

Example-2: Find a root of the equation $x \sin x + \cos x = 0$.

Solution: Given that,

$$x \sin x + \cos x = 0$$

$$\text{Let, } f(x) = x \sin x + \cos x$$

$$\begin{aligned}\therefore f'(x) &= x \cos x + \sin x - \sin x \\ &= x \cos x\end{aligned}$$

$$\begin{aligned}\text{Now, } x_{n+1} &= x_n - \frac{f(x_n)}{f'(x_n)} \\ &= x_n - \frac{x_n \sin x_n + \cos x_n}{x_n \cos x_n}\end{aligned}$$

Newton-Raphson Method (Cont'd)

Choose $x_0 = \pi$,

Step No.	x_n	$x_{n+1} = x_n - \frac{x_n \sin x_n + \cos x_n}{x_n \cos x_n}$
0	3.1416	2.8233
1	2.8233	2.7986
2	2.7986	2.7984

The difference between x_2 and x_3 is 0.0002. Hence a root, correct to three decimal places is 2.798.

N.B.: For Iteration and Newton-Raphson method, we will commonly choose $[1, \pi]$ for trigonometric ratio and $[0, 1]$ for exponential function.

Newton-Raphson Method (Cont'd)

Self Practise

Example- 2.17, 2.18, 2.19

N.B.: While computing trigonometric ratio, the specified value must be converted into degree using,

$$\text{Degree} = \frac{180 \times X}{3.14}$$

Where, X is in radian unit.

Task related to Newton-Raphson Method

1. Write some advantages of Newton-Raphson method.
2. Write some disadvantages of Newton-Raphson method.

Any Question???